Unit 2 - Computer Systems
Assignment 2 - Best Systems

Introduction

As a trainee consultant for a local company, selling and supporting computer and networking equipment, my responsibility is obtaining and designing computer systems for a range of clients. My specialisation is in photographic and graphics related equipment and software.

The manager has given a project that includes advising a client who is a photographic enthusiast with a set of requirements and a budget on buying or designing a complete computer system, including hardware, peripherals and software.

I have been asked to research and report on the following:

(P4) Recommend a computer system for a given business purpose

Task description: *You have been asked to specify a computer system (hardware and software) for a photographic enthusiast who regularly take photographs and is considering submitting work for an exhibition. You must include the detailed design choices made for type of Processor, Graphics adapter, Monitor, Memory and Storage as well as the peripherals and software they will need. There is an initial budget of £750.*

The user is expecting a professional presentation of the report and has given a preference for a word processed document with references to sources where appropriate.

Overall Description of System

The system has to be powerful enough to run photo editing softwares such as Photoshop and Lightroom well. This system is very photography focused and it's not a recommendation for a gaming or video editing PC, for those the specifications would have to be different. Since there is an initial budget of £750, the system won't be the best performing but it should be bearable for photography.

Motherboard

The motherboard that is recommended is a *Gigabyte H110M-S2H Intel H110 (Socket 1156) DDR4 Micro ATX Motherboard*. This motherboard contains 4x USB3.0, 1x PCIE x16, 2x PCI-E X1, 8x USB2.0, DDR4, 4x SATA 6Gb/s. This motherboard has a price range of £55. Refrence: www.overclockers.co.uk

Processor

The processor that is recommended is an *Intel Core i3-530 Dual-Core Processor*. This processor has a processing speed of 2.93 GHz, 2 processor cores and the processor socket is an LGA 1156. This processor has a price range of £75. Refrence: www.amazon.com

Graphics adapter

The graphics adapter that is recommended is a *NVIDIA GeForce GTX 750 TI*. This graphics adapter has 2GB of graphics card RAM, DDR5 RAM type and a HDMI hardware connectivity. This graphics adapter has a price range of £120. Refrence: www.amazon.com

Monitor

The recommended monitor is a *Monoprice 27" IPS-Zero-G Display*. This monitor has a 27” display size, IPS panel with LED backlight display type, a resolution of 2560x1440 and 60Hz refresh rate. The price range of this monitor is £200. Reference: www.monoprice.com

Memory

The recommended memory is a *Kingston KTH-PL421/8G 8GB DDR4-2133MHz ECC Memory Module*. It has a memory size of 8,192 MB, memory speed of 2133MHz, 288-pin DIMM and DDR4 SDRAM RAM type. The price of this memory stick has been reduced by 70% from a RRP of £140, giving the discounted price of £42. Reference: www.amazon.com

Storage

The recommended storage is a *Crucial MX300 525 GB SATA 2.5" 7mm (with 9.5mm adapter) Internal SSD*. It has a total capacity of 525 GB, SATA 6.0Gb/s interface, 530MB/s Read and 510MB/s Write. The price range of this storage unit is £120. Reference: uk.crucial.com

Input Devices

The recommended input devices are a *Logitech Desktop MK120 Keyboard And Mouse Set*. This bundle comes with an optical mouse with 1000 dpi resolution, a keyboard and they are both connected via USB. The price range of this kit is £14. Referece: www.uk.insight.com

Output Devices

The recommended output device is the *HP - DeskJet 1112 Printer*. This output device has a 4800 x 1200 dpi optimized colour resolution, 1200 x 1200 when printing black only, 1 USB port, 60-sheet input and 25-sheet output capacity. The price range for this printer is £25. Referece: www.bestbuy.com

Software (Operating Systems)

The recommended operating system is the *Windows 10 Home - 64-bit*. This operating system is 64-bit, includes universal apps, always up-to-date, familiar and modern interface and has the cortana AI. The price range for this operating system is £80. Refrence: www.newegg.com

Software (Applications)

The recommended application is the *Adobe Photoshop CC*. This application is the latest app from adobe, easy to use tools that helps with creating amazing image for design projects, combine multiple images, remove objects and retouch photos using a complete set of professional tools. The price for photoshop is a monthly pay of £17.15 including VAT. Refrence: www.adobe.com

(M2) Justify the choice of computer system to meet a given business purpose

Task description: *You have met with the photographic enthusiast and he asked you to justify your choice for each item explaining how these choices will meet their needs. You will need to justify each of the design choices in chosen in task 1 explaining in detail how the user will be able to utilise the advantages of that component.*

The user is expecting a professional presentation of the report and has given a preference for a word processed document with references to sources where appropriate.

Overall Description of System

The recommended computer system that has been stated meets the budget of £750, with the whole system costing £723.15. All these Specifications are meant to fit into a desktop computer, the reason that a desktop is chosen instead of a laptop is because a desktop's specifications can be replaced for newer and better specs, a laptop cannot replace its specs such as CPU or GPU and very few laptops can replace the RAM.

A PC was chosen over a MAC because a PC has smaller initial investment, more configurations and options and it's more upgradable. The PC also has its disadvantages, for example the pc is less durable (unless custom built), requires more tech support and can have more malware attacks.

Motherboard

The specific motherboard was chosen because it has a good amount of USB2.0 and 3.0 slots. USB 3.0 is good because of its speeds, it may not be as fast as apple's thunderbolt, but very few PC motherboards have thunderbolt built into them and it isn't very well supported. USB 3.0 is much faster than USB 2.0 so it's good to have a few on a PC. The motherboard also has a

fast SATA cable at 6Gb/s, 1x PCIE x16, A PCI-3 X1 and a LGA 1156 slot which is compatible with the processor.

Processor

The processor is recommended because it's part of the Intel “Core i” processors. AMD often has a better price to performance over Intel, but when it comes to running photo editing software, the software is very heavily optimized for Intel processors. Even though Photoshop and Lightroom will still run on AMD, and there won't be problems in launching the application, but Intel is worth the money more than AMD.

The least that could be recommended is a Core i3 processor, the software can run on a Pentium or Celeron processor from intel, but not as well.
The processor that has been chosen is also compatible with the motherboard because it's a LGA 1156 processor.

Graphics adapter

Nvidia graphics was recommended for the the graphics adapter, this isn't because AMD graphics adapters are “bad”. The reason is because Adobe doesn’t support anything but Nvidia on a PC. AMD is improving and it's only going to get much better in the coming years because of the AMD graphics in the Mac Pro being supported well, which should force adobe to support AMD.

For now, AMD graphics adapters should be avoided in the PC - laptop or desktop. The least that can be recommended is an Nvidia chipset graphics card with 1GB of RAM. The graphics adapter that has been chosen is compatible with the motherboard because it has a PCI express x16 slot.

Monitor

The monitor that has been recommended is a large, high resolution IPS monitor. Having this type of monitor instead of a common monitor makes a huge difference in photo editing. The monitor should at least be a 27” or 30” monitor and the resolution should range from 2560x1440 or 2560x1600 (more than 2x more resolution than HD). It would benefit more if the monitor was connected through DVI or DisplayPort, but due to the budget, a HDMI monitor had to be used with the recommended resolution and size.

Memory

The RAM that has been recommended is 8GB, which you can get by with. It’s recommended to get a larger size than 8GB, but due to the budget that wouldn't be possible. 32GB RAM is recommended because photo editing software programs are memory hogs. The software can

run pretty well with 16GB of ram and can get by with 8GB, as mentioned before, but if the budget was larger, I would max out a system configuration for a computer and get as much memory as possible.

Storage

An SSD hard drive was recommended, the reason for this is that SSD uses flash memory technology, similar to the memory in a smartphone and SD cards. They are faster at reading and writing than the magnetic spinning hard drives, but they are more expensive than HDD, but it does make a difference for photo editing.

An SSD with 256Gb or higher is recommended for the OS and programs, and to use as a “working” drive. If the budget was higher, a 1-4TB magnetic hard drive would also be added for longer-term storage, which can spin at 7200 RPM (instead of the very common 5400 RPM), which makes a big difference.

Input Devices

The input devices are the least of the priorities on building a photo editing system, but it’s still important to have a mouse and a keyboard. The only recommendation is an inexpensive, working mouse and keyboard which was listed on the system recommendations.

Output Devices

Like the input devices, output devices aren’t very important for a photo editing system. An InkJet printer which was used for the “recommended” system is good for printing out projects, but most files are stored in clouds and can be sent to other individuals online, so the printer is almost obsolete in sending photos (unless the photo is ordered via hard copy).

Software (Operating Systems)

Windows 10 Home - 64-bit was recommended because it's the latest and greatest OS from microsoft. It’s recommended to have 64-bit in the OS version or you won’t be able to use all the RAM that has been installed in the computer. Even though most PC come with 64-bit windows in the last 3-4 years, the mistake of buying a 32 bit OS when custom building a PC is a possible mistake.

Software (Applications)

Adobe’s photoshop is undoubtedly one of the most popular photo editing software today. It’s recommended because it's a very professional image processing tool, but it’s also hard to master due to its complexity. Once it is mastered the ability to manipulate photos is only limited

to the designers imagination. With many other image processing tools coming out, they could solve almost all your image problems.

(D2) Explain and justify improvements that could be made to a computer system

Task description: *The user has decided that your choice of Processor, Graphics adapter card and RAM needs to be improved. They feel the specifications of these items could be improved. They have given you an extra £250 for these improvements. You need to suggest an improved component for each function and explain why you have chosen the item against other alternatives and how it will improve the performance of the system. Finally you will need to describe the overall improvement the user will experience using these components.*

The user is expecting a professional presentation of the report and has given a preference for a word processed document with references to sources where appropriate.

Processor

With an extra budget of £250, I was able to find an improved processor that has a reduced price of %50 off. The reason this is done is because I have been asked to get a better performing processor for the build, due to the user saying that the processor is one of the specifications that need to be improved.

Identification and Specification

The improved processor is the *Intel Core i5 4460 3.20GHz Socket 1150 6MB L3 Cache Retail Boxed Processor*. This processor is a pretty big step up from the i3, due to the next gen i5 Intel family, a quad core processor instead of a dual core and a 3.2 GHz core frequency.

This processor has a 50% discount from its RRP of £175.98, giving it £87.99 discount price. Reference: www.ebuyer.com

Explanation of alternative components discounted

The alternative components was an *Intel® Core™ i7-7Y75 Processor (4M Cache, up to 3.60 GHz)* and an *Intel® Core™ i5-4460T Processor (6M Cache, up to 2.70 GHz)*. Both these processors are very good, especially the i7, and they are much better than the previously recommended i3 processor, but these weren't added the the improvement.

The reason that these weren't added is that the i7's price is significantly higher than the provided budget, making it impossible to purchase. However, the *i5-4460T 2.70 GHz* was in the price range, but it had a worse clock speed than the *i5 4460 3.20 GHz*, and the *i5 4460 3.20 GHz* had a relatively cheaper RRP price and it had a 50% discount. Refrences: ark.intel.com

Advantage that the user will experience by choosing this component.

The advantage the user will have of using the *Intel Core i5 4460 3.20GHz* over the *Intel Core i3-530 Dual-Core Processor* is that the i5 has a quad core processor, this means that the i5 can run a lot more programs at one time, or work on a large problem more efficiently than a dual core processor. This is a big improvement from the last recommended processor, the most significant being the upgrade from a dual core to a quad core processor.

The clock speed on the recommended one is also a lot higher, saying thats its faster isn't a perfect way of putting it, but explaining clock speed is a lot more complicated than it is, but having a larger clock speed is a good thing. The i5 is also a more modern processor than the i3, meaning that it can run newer programs and OS better than the i3.

RAM

The user also told me to get a different RAM for the PC build, the reason being is because it needs an improvement from the recommended RAM that was stated before. With an additional budget of £250 which already taken away by £87.99, gives me £162.01, which means that I will have to find RAM at a very low price so I could improve the other component that needs improvement.

Identification and Specification

The improved RAM is the *Kingston KTH-PL421/16G 16GB DDR4-2133MHz ECC Memory Module*. This RAM is the exact same brand and model as the last recommended one, the only difference is the the amount of RAM is doubled from 8GB to 16GB.

The price of this RAM has been reduced by 73% from its RRP of £252.28, giving a discounted price of £68.99. Reference: www.amazon.com

Explanation of alternative components discounted

The alternative components that were discounted where the *G.Skill Trident Z DDR4 RAM 32GB* and a *Corsair Vengeance 16GB (2x8GB) DDR3 1600MHz (PC3 12800) Desktop Memory*. These memory stick are very good and they are all significantly better performing than the previous recommended RAM.

However, these had to be discounted, the reason the *G.Skill Trident* was discounted was the ridiculous price of £240.29, which does fit in the budget but I wouldn't be able to purchase other component improvements for the computer build, so it had to be discounted.

The reason the Corsair *Corsair Vengeance* was discounted is that it was a DDR3 stick instead of a DDR4, this means that this isn't compatible with the motherboard. The *Corsair Vengeance*

is relatively cheap for 32GB of RAM, with a price range of £80, but due to compatibility issues with the motherboard it had to be discounted. Reference: www.amazon .com; www.handtec.co.uk

Advantage that the user will experience by choosing this component.

The advantages the user will experience is that the upgraded 16GB will be able to use multiple instances of the same bulky program, or a number of different programs simultaneously without any difficulty. Another significant reason, especially for a PC built for photography, is that photo editing software programs are memory hogs, and with 16GB of RAM, things can work pretty well and smoother than the previous 8GB of RAM.

Graphics Adapter

This is the final component that the user has recommended that I should upgrade the graphics adapter into a better one. Since the additional budget of £250 has been reduced by £156.98, giving me a final budget of £93.02.

Identification and Specification

The improved graphics adapter is the *ZOTAC GeForce GTX 1050 mini Graphics Card (ZT-P10500A-10L)*. This graphics adapter is a very modern GPU because it's part of Nvidia's new 10 series. This graphics card is 4GB graphics card RAM, 128-bit memory bus width and GDDR5 graphics card RAM type.

The price range of this graphics card is £87.32, giving the final budget of £5.70 to spare. Reference: www.amazon.com

Explanation of alternative components discounted

The alternative components that were discounted were the *Asus NVIDIA GeForce GTX 1050 EX-GTX 1050-2G 2GB GDDR5 128 Bit Memory HDMI/DP/DVI PCI Express 3 Graphics Card* and the *ASUS GTX750TI-OC-2GD5 - NVIDIA GTX 750Ti OC 1150MHz 5400MHz 2GB 128 bit DDR5 DUAL-LINK DVI-I/DVI-D HDMI PCI-E GRAPHICS CARD*.

The reason both the cards were discounted was because of the price and the budget, the *Asus NVIDIA GeForce GTX 1050* costs £146.22 and the *ASUS GTX750TI-OC* costs £129.99.

With the budget brought down to £93.02, it would be impossible to buy any of these graphics adapters, even though all these graphics adapters are fully compatible with the motherboard due to their SATA connections and they are very powerful because they are modern and they have the latest graphics RAM type of DDR5.

Advantage that the user will experience by choosing this component.

The advantage of using *ZOTAC GeForce GTX 1050 mini* over the previous graphics adapter, the *NVIDIA GeForce GTX 750 TI,* is that the 1050 is a much more modern and has one of the biggest price to performance ratio out of any graphics card to date. S

ince Adobe Photoshop supports Nvidia, the new version of photoshop which was included in the software recommendations, would support the new 10 series graphics adapters such as the 1050, which I have recommended for the improvement. Therefore, the software and OS will run a lot more smoother with the GTX 1050 than the GTX 750 TI, giving the user a better experience on running heavy programs such as photoshop.